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PMC Managers Limited — Business Analysis: Gap Assessment

Prepared for: Client / Board Discussion **Date:** February 2026 **Version:** 1.0 **Prepared by:** Delivery Team

1. Executive Summary

You have two new source documents on the table:

1. **PMC Transaction Structure & Funds-Flow Report** (v1.0, 24 July 2026) — a **commercial-legal** framework describing how money flows and who must hold what.
2. **PMC Build Sheet v2 AWS** (24 July 2026) — a **technical re-architecture** describing a native-AWS stack with four discrete applications.

Against this, we have already delivered a fully-functional **v2.0 Tender Management Platform** built on Emergent (FastAPI + MongoDB + React). It covers registration, bidding, evaluation, award, and audit — the classic tender lifecycle.

The core message for the client: > **The new scope is not an incremental change. It is a strategic pivot in three dimensions at once: architecture (Emergent stack → AWS-native), scope (single unified platform → four discrete apps), and commercial model (a fee/bid platform → a certified-instruction platform that stewards other people's money).**

The client must be made to understand these three shifts so that decisions on time, cost and re-use can be made deliberately, not by drift.

2. What We Have Already Built (Baseline)

Domain	Delivered v2.0 Feature	Status
Platform	Emergent SaaS · React SPA · FastAPI · MongoDB · single-region	Live in production
Auth	JWT (httpOnly cookies) + Emergent-managed Google OAuth	Working
Vendor onboarding	Pre-Tender Registration Form (Sections A–E), KYC uploads, 6 declarations + eSignature	Working

Domain	Delivered v2.0 Feature	Status
Stripe fee	One-time vendor registration fee via Stripe checkout	Working
Tender lifecycle	Unified RFP / RFQ / Tender with draft → pending_approval → open → closed / awarded	Working
Multi-level approval	Buyer Orgs, Departments, budget-threshold approval chains	Working
Two-envelope RFP	Sealed technical + commercial envelopes, timed open, technical qualification gate before award	Working
Bidding	Bid creation, comparison view, lowest-bid highlight, RFP weighted scoring	Working
EMD (Earnest Money Deposit)	Stripe-backed EMD, refund / forfeit workflow	Working
Pre-bid Q&A + Corrigenda	Anonymised questions, buyer answers, addenda publishing	Working
Notifications	In-app + Resend email (branded)	Working
Live feed	Server-Sent Events tender feed with filters + per-user read state	Working
Audit trail	Application-level SHA-256 hash-chained immutable ledger, admin verify button	Working
FX Conversion	Live rates (`open.er-api.com`), currency chip on tender detail	Working
SBD Template Library	Admin uploads standard bidding docs; buyers attach via dropdown	Working
Vendor KPI Dashboard	Win-rate, avg price/delivery, category history, suggestions	Working
i18n	English + Russian (Kazakh partial)	Working
Deployment	Deployed to Emergent-managed production URL	Live

Bottom line: we have a working, tested tender platform (28/28 P2 tests + 42/42 P1 tests pass) that could take live tenders tomorrow.

3. The New Scope, Decoded

3.1 Document 1 — Transaction Structure Report

This is a **commercial and legal framework**, not a build spec. It says:

- **Money splits into two categories:**
  - **Category A — PMC Revenue** (fees for services PMC performs). PMC keeps this.
  - **Category B — Stewarded Funds** (EMD, performance bonds, retention, drawdowns). PMC never holds this — it instructs and reconciles only.
- **Four money pipes**, each with a different holder:
  1. PMC fees → PMC's operating accounts via commercial gateway
  2. Bid security (EMD) → regulated escrow agent
  3. Performance guarantees + retention → guarantee bank
  4. Construction drawdowns → bond trustee / paying agent
- **Ten roles to appoint** externally: Escrow Agent, Guarantee Bank, Bond Trustee, Security Agent, Paying Agent, Independent Certifier, Account Bank, Payment Gateway, Legal Counsel, KYC/Screening Provider.
- **Six controls to build in**: segregation, account-control agreements, certified-instruction workflow, float-interest disclosure, immutable audit trail, related-party register.
- **Six open legal questions** blocking activation of the corresponding money flows.

Implication for us: This is largely **legal & organisational work outside the platform**, but it imposes concrete technical requirements — most notably a **certified-instruction workflow** and **immutable ledger** that must interface with regulated third parties.

3.2 Document 2 — Build Sheet v2 AWS

This is a **complete technical re-architecture**. It says:

- **Move off Emergent's managed stack** to AWS-native: RDS PostgreSQL 16 Multi-AZ · S3 with Object Lock (Compliance Mode, 7-year retention) · Cognito · ECS Fargate · SES · CloudWatch · Terraform IaC.
- **Split the platform into FOUR separate applications**:
  1. **App 1 — Milestone Evidence Capture** (React Native mobile, offline-first with SQLite, GPS-stamped, witness-signed, WORM evidence).
  2. **App 2 — Package Tracker** (internal, Retool, admin/site_lead only).
  3. **App 3 — Pre-Tender Registration Portal** (React web, magic-link resume, EN/RU/KK).
  4. **App 4 — Public Tender Notice Board** (Next.js SSR, SEO-optimised, no login).

- **One central FastAPI + SQLAlchemy API** on Fargate — apps never touch the database directly.
- **Multi-tenancy via PostgreSQL RLS** driven by session variables from Cognito claims.
- **Explicit OUT-OF-SCOPE clause**: "never implement — encryption, key management, **bid submission, bid opening, evaluation scoring, contract award, payment processing**."

Implication for us: The document deliberately **omits most of what we already built** and replaces it with **a milestone-evidence and public-notice-board focus**. Bidding, evaluation and payment are explicitly excluded from this build phase.

4. Gap Analysis — Point-by-Point Comparison

#	Area	Current (v2.0, delivered)	New Scope (Doc 1 + Doc 2)	Gap Type	Impact
1	**Cloud platform**	Emergent-managed (opinionated PaaS)	AWS-native (RDS · S3 · Cognito · Fargate · Terraform)	**Replacement**	HIGH — full re-platform
2	**Database**	MongoDB (document, single-tenant)	PostgreSQL 16 Multi-AZ with RLS multi-tenancy	**Replacement**	HIGH — schema + query rewrite
3	**Auth**	JWT + Emergent Google OAuth	AWS Cognito with hosted UI + MFA optional / admin mandatory	**Replacement**	MEDIUM — user & session migration
4	**Tenancy model**	Single-tenant (implicit)	Multi-tenant via `tenant_id` from Cognito, enforced by RLS `force row level security`	**Addition**	HIGH — data model change
5	**File storage**	MongoDB blob attachments +	S3 with SSE-KMS + Object Lock (Compliance Mode, 7-year, WORM) —	**Replacement + Upgrade**	HIGH — evidence

#	Area	Current (v2.0, delivered)	New Scope (Doc 1 + Doc 2)	Gap Type	Impact
		Emergent object store	**irreversible on day one**		chain-of-custody
6	**Tender lifecycle (draft → open → award)**	Full lifecycle implemented	**Out of scope** per new build sheet Part 5	**Removal from Phase 1**	CRITICAL — clarify with client
7	**Bidding, evaluation, scoring**	Full RFP/RFQ bidding, weighted scoring, comparison, award	**“Never implement”** in the new phase	**Removal from Phase 1**	CRITICAL — clarify with client
8	**Two-envelope sealed bidding**	Delivered	Not in scope	**Removal from Phase 1**	Retain code for later phase
9	**EMD workflow**	Stripe-backed EMD refund/forfeit	Doc 1 mandates EMD sits in a **regulated escrow account**, not with PMC. Doc 2 does not implement it.	**Redesign**	Existing EMD code is incompatible with Doc 1’s regulatory model
10	**Payment processing**	Stripe for registration fee + EMD	**“Never implement”** in the new phase	**Removal from Phase 1**	Keep Stripe route but scope out
11	**Vendor pre-tender registration**	Delivered (Sections A–E, single app)	App 3 (fresh build, magic-link resume, EN/RU/KK)	**Rebuild in new stack**	Existing form logic re-usable as spec
12	**Public tender notice board**	Not built (auth required for tender views)	App 4 (Next.js SSR, no-auth, SEO + JSON-LD)	**Addition**	New workstream

#	Area	Current (v2.0, delivered)	New Scope (Doc 1 + Doc 2)	Gap Type	Impact
13	**Milestone evidence capture**	Not built	App 1 (mobile, offline-first, WORM, GPS, witness signature, idempotent sync)	**Addition**	Largest new workstream
14	**Package Tracker (internal)**	Partial (admin dashboards)	App 2 (Retool, drag-and-drop board, 203 KZ packages import)	**Replacement / Rebuild**	Use Retool = faster
15	**Multi-language**	EN + RU + partial KK	EN + RU on App 1 & App 4; EN + RU + KK on App 3	**Extend**	Add KK completeness on registration app
16	**Audit trail**	Application-level SHA-256 hash chain in MongoDB	**S3 Object Lock (Compliance Mode)** for evidence + audit_event table with append-only triggers	**Upgrade**	New scope's WORM is stronger — regulator/underwriter-grade
17	**Certified-instruction workflow**	Not built explicitly	Required by Doc 1 Section 7 for every release of EMD, retention, drawdown	**Addition**	New workstream
18	**Regulated third-party integration**	None	Doc 1: Escrow Agent, Guarantee Bank, Bond Trustee, Paying Agent,	**Addition**	Depends on legal appointments

#	Area	Current (v2.0, delivered)	New Scope (Doc 1 + Doc 2)	Gap Type	Impact (Doc 1 Section 6)
			Account Bank must all be integrated for instructions & reconciliation		
19	**Independent Certifier / Engineer role**	Not built	Doc 1 positions PMC as Independent Certifier via the milestone evidence engine (App 1)	**Addition**	App 1 is the deliverable that makes this role real
20	**Related-party register + float-interest disclosure**	Not built	Doc 1 Section 7 controls	**Addition**	Small module
21	**Deployment / IaC**	Emergent one-click	Terraform in `pmc-infra` repo, five GitHub repos, branch protection, CI scans	**Replacement**	DevOps re-work
22	**Data residency**	Emergent default region	Native regional (Kazakhstan) as a Terraform variable	**Upgrade**	Enables gov / in-country deals
23	**Monthly OpEx**	Emergent subscription	~$450–525 / month (RDS + Fargate + S3 + SES + Cognito + Retool + Emergent build months)	**Increase**	+~$150/month over shortcut route

5. What the Client Must Decide

There are **five decision points** that will determine the delivery plan and budget. None of these can be answered by us alone.

Decision 1 — Do we replatform, or do we extend?

- **Option A (Replatform to AWS as Doc 2 says):** Rebuild against AWS-native stack. The v2.0 delivered platform becomes a reference/spec, not the running system. Estimated effort **7 weeks minimum** per Doc 2 Part 8.
- **Option B (Extend the current Emergent platform):** Add App 1 (mobile evidence capture) and App 4 (public board) alongside the existing platform on Emergent. Faster, cheaper, but does not deliver Object Lock, native residency or the multi-tenant AWS control plane.
- **Option C (Hybrid):** Keep v2.0 running for tender lifecycle (which the new build sheet explicitly excludes), and build Doc 2's four apps in AWS in parallel. Costs and integration complexity go up but no functionality is lost.

Our recommendation: Option C, phased — because the delivered tender lifecycle is production-ready and the new build sheet does not replace it, only adds to it.

Decision 2 — Is bidding really out of scope, or only out of Phase 1?

Doc 2 Part 5 says "**Never implement — bid submission, bid opening, evaluation scoring, contract award, payment processing.**" But those features are **already built and working** in v2.0. This looks like a **phasing statement** ("do not build in the new AWS apps yet") rather than an instruction to **delete** them.

Client must confirm: - Does "never implement" mean (a) don't touch it in the new AWS apps, keep v2.0 running, or (b) formally decommission bidding entirely?

Decision 3 — Who holds the money?

Doc 1 makes it clear that PMC **must not** hold EMD, performance bonds, retention or construction drawdowns. Today, our v2.0 Stripe-backed EMD workflow does exactly that (funds sit in Stripe under PMC's account).

Client must confirm: - Which regulated party will hold EMD? (Kazakh bank escrow division? AIFC escrow company? A DIFC / ADGM specialist?) - Until this is decided, the certified-instruction workflow cannot be turned on for real money — but the platform can still be built to instruct-and-reconcile against a mock/test escrow.

Decision 4 — Open legal questions (Doc 1 Section 9)

Six questions gate the corresponding money flows. Until each is answered, that flow stays disabled in the platform:

1. Land-grant procurement obligations?
2. Concession / PPP status?
3. Legality of verification / document fees?

4. Who may hold EMD and guarantees?
5. Float-interest entitlement?
6. Bond-terms disclosure obligations?

Client must engage counsel in UK, Kazakhstan and Luxembourg **before Phase-1 go-live**.

Decision 5 — GitHub org & AWS account ownership

Doc 2 assumes a **`pmc-managers` GitHub org with five repos** and a dedicated AWS account with Terraform-managed infra. Both must exist and be owned by the correct legal entity before we can begin.

Client must: - Create/nominate the AWS account (payer, region, tag policy). - Create/nominate the GitHub org with billing plan. - Confirm who owns the Terraform state (recommend an S3 backend with lock table in the same AWS account).

6. Proposed Phased Roadmap

Assuming the client agrees to Option C (hybrid — extend + build new apps in AWS):

Phase 0 — Governance & Legal (Weeks 1–4, in parallel)

- Counsel engaged on 6 open legal questions
- Escrow Agent, Guarantee Bank, Bond Trustee identified & LOI'd
- AWS account + GitHub org provisioned
- Terraform bootstrap (Part 1 of Doc 2)

Phase 1 — AWS Foundation (Weeks 2–3)

- Terraform provisions RDS, S3 (Object Lock), Cognito, ECS, SES, Secrets Manager, CloudWatch
- **Object Lock configured at bucket creation** (Doc 2 warns this is irreversible)
- `pmc-api` FastAPI + SQLAlchemy scaffolded on Fargate
- RLS policies + tenant middleware working (RLS proof-test 5, 7, 8, 9 in Doc 2 Part 7)

Phase 2 — App 4 Public Notice Board (Week 4)

- Next.js SSR, no auth, SEO + JSON-LD
- Reuses existing v2.0 tender data via a read-only sync/export bridge until full migration

Phase 3 — App 2 Package Tracker in Retool (Weeks 4–5)

- 203 KZ packages CSV imported
- Contract & milestone setup (this is the data App 1 consumes)

Phase 4 — App 1 Milestone Evidence Capture (Weeks 5–7)

- React Native mobile
- SQLite offline queue
- Presign → S3 → sync flow (idempotency, immutability, Object Lock proofs)

- Two site-device pilot

Phase 5 — App 3 Pre-Tender Registration Portal (Weeks 7–12)

- Rebuild vendor registration in the new stack (reuses v2.0 form logic as spec)
- EN/RU/KK
- Magic-link resume
- Approval workflow ties into existing v2.0 admin queue via API bridge, or fully migrates

Phase 6 — Certified-instruction & Escrow Bridge (Weeks 10+)

- Machine-readable release instructions to appointed escrow agent
- Reconciliation dashboards
- Related-party register + float-interest disclosure UI

Phase 7 — v2.0 Bidding Platform Decision (post-Phase 5)

- Depending on Client decision, either:
  - Migrate v2.0 bidding module onto AWS stack, OR
  - Keep v2.0 on Emergent as the bidding platform and integrate via API, OR
  - Decommission v2.0 bidding

7. Risks & Considerations

Risk	Severity	Mitigation
Object Lock is irreversible — a wrong config on day one means rebuilding the bucket later and losing chain-of-custody on everything already in it (Doc 2 warns explicitly)	HIGH	Dry-run in a non-prod bucket first; run the Object-Lock refusal test (Part 7 item 9) before any real evidence uploaded
RLS silent failure — `set local` outside a transaction returns zero rows silently (Doc 2 4.1)	HIGH	Explicit failure-mode test (Part 7 item 5) written into CI
v2.0 investment write-off — if Option A (full replatform) is chosen, ~4 months of v2.0 work becomes reference material, not running code	HIGH (financial)	Reuse spec, form structure, UX patterns, evaluation logic as documentation for the AWS rebuild
Regulatory delay — 6 open legal questions could delay real-money go-live by 3–6 months	HIGH	Build against **mock escrow** endpoints so the platform is ready before the appointments are finalised
Multi-tenancy retrofit — v2.0 is single-tenant. AWS build is multi-tenant with	MEDIUM	Snapshot v2.0 as tenant #1 during migration

Risk	Severity	Mitigation
`tenant_id` everywhere. Migrating existing data means synthesising a tenant_id per record.		
Cognito ≠ Emergent Google OAuth — existing users have Emergent-managed identities. Migration to Cognito is a one-way door for existing session state.	MEDIUM	Password reset flow for all existing users at cutover; SES email templates prepared
Stripe out-of-scope — Doc 2 Part 5 explicitly excludes payment processing. Existing Stripe integration must be retired or moved out of the AWS build.	MEDIUM	Retain Stripe only in v2.0; new AWS build never touches payments (per doc)
Kazakh + Russian + English language load — App 3 requires KK; current i18n is EN/RU + partial KK	LOW	Complete KK strings during App 3 build
GitHub push blocker (current)	LOW	Resolve via OAuth flow (see prior guidance) before Phase 0

8. Cost & Timeline Summary

Item	Baseline (delivered v2.0)	New Scope (Option C hybrid)
Build effort remaining	~0 (production-ready)	~10–12 weeks across 4 apps + API + infra
Monthly OpEx	Emergent subscription only	~$450–525 / month (AWS) + Emergent build months
Regulatory / Legal cost	Not applicable	3-jurisdiction counsel engagement — external estimate needed
Third-party appointments	Not applicable	Escrow agent, guarantee bank, trustee, paying agent — negotiate individually
Risk of write-off	Delivered, protected	HIGH if full-replatform is chosen without hybrid

9. Recommended Client Conversation Agenda

Use this in your meeting.

Slide 1 — What we already built (5 min) — Show v2.0 dashboard, tender lifecycle, KPI, audit trail. Establish the baseline.

Slide 2 — The two new documents (5 min) — One is legal-commercial, one is technical. They point in the same direction but sit at different layers.

Slide 3 — The three shifts (10 min) — Architecture · Scope · Commercial model. Confirm the client actually intends all three.

Slide 4 — The five decisions we need from you (15 min) — Walk through Section 5 of this doc. Get answers on the record.

Slide 5 — The proposed hybrid roadmap (10 min) — Section 6. Confirm phasing.

Slide 6 — Risks & irreversibility warnings (5 min) — Section 7, especially Object Lock and the 6 legal questions.

Slide 7 — Sign-off & next steps (10 min) — What we start next week; what the client must produce (AWS account, GitHub org, counsel engagement).

10. Appendix — Terminology Cheat Sheet for Non-Technical Stakeholders

Term	Plain English
Category A / B money	Money PMC earns (keep it) vs money PMC just watches over (never hold it)
Escrow	A regulated bank account run by a neutral third party that holds funds until a defined event
EMD	Bid deposit — bidders lodge it to prove they are serious; refunded if they lose
Retention	Money held back from a contractor's payment until they fix any defects after handover
Certified-instruction workflow	An audit-proof approval process — "no money moves unless the system says so"
Object Lock (Compliance Mode)	AWS S3 feature — once a file is written, nobody can delete or change it (not admins, not the CEO) for the retention period. Regulator-grade evidence.
Row-Level Security (RLS)	Database-level rule that ensures user X can only see their own tenant's rows — even if the app has a bug

Term	Plain English
RLS + `force row level security`	Even the table owner respects the rule. A safety net against admin mistakes.
Multi-tenant	One platform, many client organisations, strictly isolated from each other
WORM	Write-Once-Read-Many — evidence that cannot be tampered with after creation
Idempotent sync	Send the same batch 10 times, still creates one record. Safe for flaky mobile connections.
Presigned URL	A time-limited link that lets a mobile app upload directly to S3 without the server acting as a middleman
SSR (Server-Side Rendering)	Pages rendered on the server first, so Google can index them (matters for the public notice board)
Bond trustee / paying agent	The professionally-appointed institutions that hold and disburse the bond money (never PMC)
Independent Certifier	The party who signs off “yes, this milestone is truly complete” — the platform’s evidence engine positions PMC for this role
Related-party	A counterparty connected to the sponsor. Not forbidden, but must be arm’s-length and disclosed.

This document is a delivery-team analysis for client-facing discussion. It is not legal advice. All legal questions in Doc 1 Section 9 must be answered by qualified counsel before the corresponding money flows are enabled.